

A fixed point defeats the proposed periodic-height gap

Disproof of conjecture 00000007777

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Conjecture 00000007777 defines m_n as the minimum canonical height of points of exact period n for a non-postcritically-finite (non-PCF) polynomial f of degree d . It asserts a positive lower bound $m_n \geq cd^n/n$, where $c > 0$ depends only on f . We disprove this inequality already at $n = 1$.

Take the quadratic polynomial

$$f(X) = X^2 + 6X \in \mathbb{Q}[X] \subset \overline{\mathbb{Q}}[X].$$

Its derivative is $2X + 6$, so -3 is a critical point. Its orbit begins

$$-3 \mapsto -9 \mapsto 27 \mapsto 891 \mapsto \dots$$

For every rational $x \geq 27$, $f(x) - x = x(x + 5) > 0$. Hence, from the third displayed value onward, the orbit is strictly increasing and remains at least 27. The first two values are distinct and negative. There is no repetition in the critical orbit, so f is non-PCF even when critical points and orbits are considered in $\overline{\mathbb{Q}}$.

Meanwhile, $f(0) = 0$, so 0 has exact period 1. For $x \in \mathbb{Q}$ in lowest terms, write its logarithmic height as

$$h(x) = \log \max\{|\text{num}(x)|, \text{den}(x)\}, \quad \widehat{h}_f(x) = \lim_{k \rightarrow \infty} 2^{-k} h(f^{\circ k}(x)).$$

Since every iterate of 0 is 0 and $h(0) = 0$, the defining sequence is identically zero. Thus $\widehat{h}_f(0) = 0$. The asserted bound for m_1 would imply $2c \leq \widehat{h}_f(0) = 0$, contradicting $c > 0$.

Lean correspondence. The Lean project defines $K = \text{AlgebraicClosure}(\mathbb{Q})$ and the actual polynomial $P \in K[X]$. It proves $\deg P = 2$ and verifies that -3 is a derivative root in K . A strictly increasing rational orbit is transported through the injective map $\mathbb{Q} \hookrightarrow K$, proving that P is non-PCF. It defines rational logarithmic height by the formula above and canonical height by the degree-normalized orbit limit; at 0 the limit is proved to equal zero. The final theorem `TLMC7777.counterexample` combines these facts and denies every positive lower bound at period 1. Its only axiom dependencies are `propext`, `Classical.choice`, and `Quot.sound`.

The argument refutes the asserted positive inequality, so the proposed optimality statement and Böttcher-series explanation require no separate analysis.